Lecture 4 Solving Separable First Order ODEs

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Engineering Mathematics III

ENGE702

Lecture 4: Solving first order separable differential equations

3 methods to analyse ODEs
① exact solution
② direction field
③ numerical method

First order ODEs

In this lecture we will see how to find exact solutions to certain classes of differential equations.

The **order** of a differential equation is the level of the highest derivative in the equation. For example:

$cos(x)y'' + x^3y^5y' = 2$ has order 2 (second order),

$x^7y' + 2y^2 = \sin(y)$ has order 1 (first order),

$y^4 - 2y'' + \sin(y''''') = 2x$ has order 5 (fifth order),

etc.....

In this lecture we will see how to solve certain first order ODEs. Later we will see how to solve a small class of second order ODEs.

Separable ODEs

Before we start on some more general classes, we will see how to solve the most basic first order equations.

Question: What is a function y such that $y = y'$?

Exercise Based on this, come up with a function y such that $ky = y'$ for a constant k.

Exercise Based on this, come up with a solution to the differential equation $y' + ky = 0$.

Separable ODEs

The first class of differential equations we will see how to solve is the class of separable ODEs.

A first order ODE is **separable** if it can be written in the form

$$g(y)y' = f(x)$$

A separable first order ODE has the form

$$g(y)\,y' = f(x)$$

$$\Rightarrow g(y)\frac{dy}{dx} = f(x)$$

$$\Rightarrow \underbrace{g(y)\,dy}_{\text{dependent variable on one side}} = \underbrace{f(x)dx}_{\text{independent variable on the other side.}}$$

Separable ODEs

We can (often) solve a separable ODE as follows:

Remember that y' is another notation for $\frac{dy}{dx}$. Knowing this, we can rearrange our ODE to be

$$g(y)\,dy = f(x)\,dx$$

From there we can integrate both sides (if possible)

Then find antiderivatives in both sides:

$$\int g(y)\,dy = \int f(x)dx$$

Example: Solve $y'+2xy=0$

$y'+2xy=0$

rearrange our ODE to be

$$g(y)\,dy = f(x)\,dx$$

From there we can integrate both sides (if possible)

$$\int g(y)\,dy = \int f(x)\,dx$$

From here, we can find an explicit solution, ie $y = h(x)$, or an implicit solution $H(x, y) = 0$.

Separable ODEs

Example: Find the general solution to the differential equation

$$y' + 2xy = 0$$

Note that we can rearrange this to be

$$\frac{dy}{dx} = -2xy$$
$$\frac{dy}{y} = -2x\,dx$$

From here we can integrate both sides and get

$$\ln(y) = -x^2 + c$$

Taking both sides to the power of e:

$$y = e^{-x^2+c}$$

or alternatively

$$y = Ce^{-x^2}$$

Example: Solve $y' + 2xy = 0$

$y' + 2xy = 0$

$\Rightarrow y' = \frac{dy}{dx} = -2xy$

$\Rightarrow \frac{1}{y}\,dy = -2x\,dx$

$\Rightarrow \int \frac{1}{y}\,dy = \int -2x\,dx$ find antiderivatives

$\Rightarrow \ln|y| = -x^2 + C$ → implicit solution. A function related to x and y implictly

$\Rightarrow e^{\ln|y|} = e^{-x^2+C}$

$\Rightarrow y = e^{-x^2+C} = e^{-x^2}\cdot e^{C} = C\cdot e^{-x^2}$ → explicit solution

Separable ODEs

Now we will look at another example, this time a more practical application.

Example: Newton's law of cooling states that the temperature $T(t)$ of a cooling object changes at a rate proportional to the difference between its temperature and the ambient temperature T_A, ie

$$\frac{dT}{dt} = k(T - T_A)$$

Suppose a cup of coffee has a temperature of $75°$, and 10 minutes later it is $65°$. The ambient temperature is $20°$.

1. Find a formula for the temperature T as a function of time t.
2. How long before the coffee cools to $50°$?

Example: $T(0) = 75°$ $T(10) = 65°$

$T_A = 20°$

Newton's law of cooling gives the ODE

$$\frac{dT}{dt} = k(T - T_A)$$

Find a general solution to the ODE:

Separate dependent variable T and independent variable t

$$\frac{dT}{T - T_A} = k\,dt$$

$\Rightarrow \int \frac{1}{T-T_A}\,dT = \int k\,dt \Rightarrow \ln|T - T_A| = kt + C$

$\Rightarrow e^{\ln|T-T_A|} = e^{kt+C}$

$\Rightarrow T - T_A = e^{C}\cdot e^{kt}$

$\Rightarrow T(t) = C\cdot e^{kt} + T_A,\ C = e^{C}$

explicit general solution

At $t = 0$,

$T(0) = C\cdot e^{0} + T_A = C + 20 = 75$

$\Rightarrow C = 55°$

Then we use $t = 10$ to get

$T(10) = 55\cdot e^{10k} + 20 = 65$

$e^{10k} = \frac{45}{55}$

$k = \frac{1}{10}\ln\frac{45}{55} \approx -0.02$

So the temperature at time t is

$T(t) = 55\cdot e^{-0.02t} + 20$

explicit particular solution

Find out how long it will take to cool down to $50°$ by solving

$T(t) = 55e^{-0.02t} + 20 = 50$

$e^{-0.02t} = \frac{30}{55} \Rightarrow t = \frac{1}{-0.02}\ln\left(\frac{30}{55}\right)$

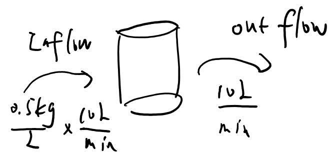
≈ 30.2 minutes

Separable ODEs

Example: Mixing problem:

A tank contains 1000 litres of water with 10kg of salt disolved into it. Brine (water with salt dissolved) containing 0.5kg of salt per litre is added to the tank at a rate of 10 litres per minute. This is mixed uniformly into the solution in the tank. The solution in the tank is also removed at a rate of 10 litres per minute.

Find a function that describes the concentration of salt in the tank at any time t.



use $y(t)$ to denote the mass of salt at time t.

$y(0) = 10$ kg

Write down an ODE for the rate of change of the salt.

$$\frac{dy}{dt} = \underbrace{0.5\times 10}_{\text{inflow}} - \underbrace{\frac{y}{1000}\times 10}_{\text{outflow}} = 5 - 0.01y$$
$$= 0.01(500 - y)$$

$\Rightarrow \frac{1}{500-y}\,dy = 0.01\,dt$

$\Rightarrow \int \frac{1}{500-y}\,dy = \int 0.01\,dt$

$\Rightarrow -\ln|500 - y| = 0.01t + C$

$\Rightarrow 500 - y = e^{-0.01t - C}$

At $t = 0$, we know

$y(0) = 500 - C\cdot e^{0} = 500 - C = 10$

$\Rightarrow C = 490$.

So the particular solution is

Text Book References

Chapter 1.3, 1.5

$$\Rightarrow -\ln|500-y| = 0.01t + C$$

$$\Rightarrow 500-y = e^{-0.01t-C}$$

$$\Rightarrow y(t) = 500 - C\cdot e^{-0.01t}$$

$$\Rightarrow C = 490.$$

So the particular solution is

$$y(t) = 500 - 490e^{-0.01t}.$$